Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

- Channel attention;
- Spatial attention;
- Sequential channel–spatial attention;
- Lightweight self-attention;
- Different positions for inserting attention within ResNet50;
- Single-stage and multi-stage attention;
- Attention ordering;
- Additional established attention mechanisms; and
- Qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

1. Explain the stage-wise architecture of a pretrained ResNet50;
2. Distinguish early, intermediate, and late convolutional features;
3. Explain the motivation behind attention mechanisms;
4. Implement channel attention;
5. Implement spatial attention;
6. Implement sequential channel–spatial attention;
7. Implement a lightweight self-attention mechanism;
8. Insert an attention module at different stages of a pretrained network;
9. Analyze the influence of attention position;
10. Compare different attention mechanisms under a controlled experimental protocol;
11. Analyze classification accuracy and class-wise performance;
12. Visualize internal attention maps;
13. Generate Grad-CAM explanations;
14. Distinguish an internal attention map from a post-hoc explanation map;
15. Analyze accuracy–complexity trade-offs; and
16. Perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.

Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.

This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

The dataset has 3670 images. The split gave 2569 training, 550 validation and 551 test images, and the same split was used for every run. Dandelion has the most images (898) and daisy the fewest (633), so the classes are only mildly imbalanced. Macro F1 is still reported next to accuracy so that the smaller classes count equally.

Table 1: Class-wise image counts in the fixed split.

Class	**Train**	**Validation**	**Test**	**Total**
dandelion	629	134	135	898
daisy	443	95	95	633
tulips	559	120	120	799
sunflowers	489	105	105	699
roses	449	96	96	641
Total	2569	550	551	3670

The images differ a lot in scale, background and lighting. Some flowers fill the whole frame while others are small with grass or leaves around them. Tulips and roses are both red or pink in many of the samples, so they are expected to be the hardest pair, and the confusion matrices later confirm this.

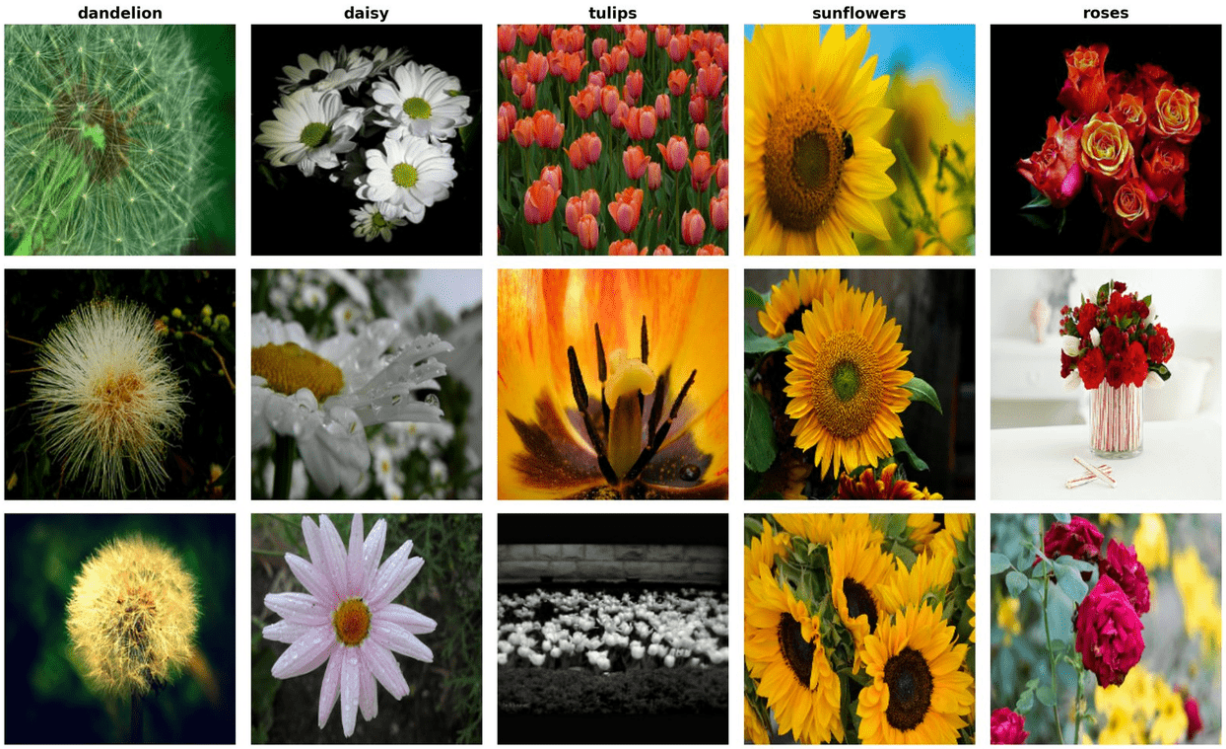


Figure 1: Sample training images from the five classes (columns: dandelion, daisy, tulips, sunflowers, roses).

4. Source Code

The complete code (split, models, training, evaluation, plots and tables) is available at:
https://github.com/Ankith-Davey/Deep-Learning-Lab-Ankith-24011101004/tree/main/Lab%208%20%26%209

5. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	Spatial Resolution	Channels
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

6. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

7. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

8. Candidate Attention Positions

$$P_1 : \text{ after conv2_x},$$

$$P_2 : \text{ after conv3_x},$$

$$P_3 : \text{ after conv4_x},$$

$$P_4 : \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

9. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

10. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^{C}.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

11. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\mathrm{Conv}_{7\times7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

12. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

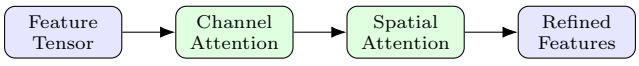
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



13. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H\times W\times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N\times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

14. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.
The attention modules did not change the feature size at any position, so the next ResNet50 stage gets the shape it expects and its pretrained weights still apply. The parameters added by Channel+Spatial grow with the channel count, from 8563 at P_1 to 526563 at P_4. This is about four times more each time the channels double.

Table 2: Input and output feature dimensions of the attention block at each position.

Position	**Location**	**Input** (H, W, C)	**Output** (H, W, C)
P_1	after conv2_x	$(56, 56, 256)$	$(56, 56, 256)$
P_2	after conv3_x	$(28, 28, 512)$	$(28, 28, 512)$
P_3	after conv4_x	$(14, 14, 1024)$	$(14, 14, 1024)$
P_4	after conv5_x	$(7, 7, 2048)$	$(7, 7, 2048)$

15. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Table 3: Attention-type ablation at P_3 (after conv4_x). Inference time is in ms per image (batch of 32).

Model	**Position**	**Train Acc (%)**	**Val Acc (%)**	**Test Acc (%)**	**Precision (%)**	**Recall (%)**	**Macro F1 (%)**	**Parameters**	**Inf. Time (ms)**
A0: Baseline	–	99.42	93.27	92.74	93.08	92.47	92.50	23 597 957	4.55
A1: Channel	P3	99.07	93.45	92.56	92.81	92.44	92.44	23 730 117	4.47
A2: Spatial	P3	98.91	93.82	91.65	91.87	91.25	91.51	23 598 056	4.50
A3: Channel+Spatial	P3	98.95	91.82	91.11	91.98	90.87	90.91	23 730 216	4.63
A4: Self-Attention	P3	98.37	93.27	91.83	91.94	91.53	91.68	24 908 678	4.73
A5: Combined	P3	98.40	93.82	92.20	92.57	91.83	92.13	25 040 937	4.76

The baseline gets 92.74% and no attention model beats it. All six models lie within 9 test images (1.6 points), so they are not ranked. A1 comes closest at 92.56%, just one image behind. A3 and A5 start slowly (44% training accuracy in epoch 1 against 70.5%), probably because two random gates scale the features down.

16. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	**Attention Position**	**Resolution**	**Channels**
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

The principal question is:

How does the location of attention influence performance?

Table 4: Attention-position ablation with Channel→Spatial attention (B3 is the same run as A3).

Model	Position	Train Acc (%)	Val Acc (%)	Test Acc (%)	Precision (%)	Recall (%)	Macro F1 (%)	Parameters	Inf. Time (ms)
Baseline	–	99.42	93.27	92.74	93.08	92.47	92.50	23 597 957	4.55
B1: after conv2_x	P1	98.05	92.73	90.93	91.26	90.39	90.71	23 606 520	4.69
B2: after conv3_x	P2	98.17	92.55	91.83	92.33	91.13	91.53	23 631 368	4.61
B3: after conv4_x	P3	98.95	91.82	91.11	91.98	90.87	90.91	23 730 216	4.63
B4: after conv5_x	P4	98.48	92.91	91.83	92.30	91.64	91.80	24 124 520	4.63

All four positions are below the baseline (90.93% to 91.83%) and lie within 5 images of each other, so the position cannot be ranked. The added parameters grow with the channels, from 8563 at P_1 to 526563 at P_4. P_1 is the lowest and also a bit slower (4.69 ms) despite the fewest extra parameters, probably due to the large 56×56 map.

17. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

Attention only at P_3,

Attention only at P_4,

and

Attention at $P_3 + P_4$.

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Table 5: Attention at one or several positions (Channel→Spatial at every position).

Model	Position	Train Acc (%)	Val Acc (%)	Test Acc (%)	Precision (%)	Recall (%)	Macro F1 (%)	Parameters	Inf. Time (ms)
Baseline	–	99.42	93.27	92.74	93.08	92.47	92.50	23 597 957	4.55
P_3 only	P3	98.95	91.82	91.11	91.98	90.87	90.91	23 730 216	4.63
P_4 only	P4	98.48	92.91	91.83	92.30	91.64	91.80	24 124 520	4.63
$P_3 + P_4$	P3+P4	98.21	94.00	92.56	92.50	92.29	92.35	24 256 779	4.66
$P_2 + P_3 + P_4$	P2+P3+P4	98.64	93.09	91.47	92.13	90.68	91.16	24 290 190	4.72

$P_3 + P_4$ gives 92.56%, better than P_3 or P_4 alone but still below the baseline. Adding P_2 drops it to 91.47%. So more locations are not always better, and the extra parameters buy nothing. $P_3 + P_4$ does have the best validation accuracy in this table (94.00%) and is only one test image behind the baseline.

Training for 20 epochs took 7.81 min for the baseline, 8.03 min for P_3, 7.89 min for P_4, 8.06 min for $P_3 + P_4$ and 8.36 min for $P_2 + P_3 + P_4$. So even three modules add only about half a minute (7%), and memory use was not recorded. The extra cost is small, but with no accuracy gain it is still not worth it.

18. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

Channel → Spatial

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Table 6: Attention order at P_3.

Model	Position	Train Acc (%)	Val Acc (%)	Test Acc (%)	Precision (%)	Recall (%)	Macro F1 (%)	Parameters	Inf. Time (ms)
Channel→Spatial	P3	98.95	91.82	91.11	91.98	90.87	90.91	23 730 216	4.63
Spatial→Channel	P3	98.13	94.91	94.01	93.96	93.86	93.88	23 730 216	4.61

Spatial then Channel gets 94.01% against 91.11% for Channel then Spatial, which is 16 images, with the same parameters. But a gap like this also appears between kernel sizes, so it is not reliable evidence for the order. It is the best of all sixteen runs (94.91% validation too), but with one seed it may just be a lucky run.

19. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3	90.74%	90.61%	4.64 ms
5×5	93.28%	93.13%	4.56 ms
7×7	91.11%	90.91%	4.63 ms

The accuracy is 90.74%, 93.28% and 91.11% for the 3×3, 5×5 and 7×7 kernels. There is no steady trend, so a larger kernel does not give a clear gain. The kernel barely changes the cost (19, 51 and 99 weights, 4.56 to 4.64 ms), and the 3×3 model also has the lowest training accuracy (97.39%).

20. Additional Task: Comparison of Different Attention Mechanisms

20.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention	92.74%	93.08%	92.47%	92.50%	23 597 957	4.55 ms
Channel / SE	92.56%	92.81%	92.44%	92.44%	23 730 117	4.47 ms
Spatial Attention	91.65%	91.87%	91.25%	91.51%	23 598 056	4.50 ms
ECA	92.56%	92.63%	92.42%	92.35%	23 597 962	4.58 ms
CBAM	91.65%	91.75%	91.62%	91.46%	23 730 216	4.63 ms
Self-Attention	91.83%	91.94%	91.53%	91.68%	24 908 678	4.73 ms
Additional Mechanism: Channel+Spatial (A3)	91.11%	91.98%	90.87%	90.91%	23 730 216	4.63 ms

SE is the same run as A1, and the last row is A3. CBAM and A3 have equal parameters because CBAM shares one channel network. Coordinate Attention was not run, as ECA and CBAM already give the two extra mechanisms. ECA adds only 5 parameters because its channel weights come from one small 1D convolution.

20.2 Questions for Attention-Mechanism Comparison

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

1. **Accuracy.** It ranges from 91.11% (Channel+Spatial) to 92.74% (no attention). SE and ECA are equal at 92.56%, and no mechanism beats the baseline. The two that use only channel weighting (SE, ECA) do a little better than the ones with a spatial part, but the whole spread is just 9 images.
2. **Macro F1.** It follows the same order, from 90.91% (Channel+Spatial) to 92.50% (baseline). SE gives 92.44% and ECA 92.35%. The gap between accuracy and F1 is only 0.1 to 0.25 points for every mechanism, so none of them is much worse on one particular class.
3. **Parameters.** The added parameters are 99 for spatial, 5 for ECA, 132160 for SE, 132259 for Channel+Spatial and CBAM, and 1310721 for self-attention. Self-attention adds about ten times more than SE, mostly because its value and output 1×1 convolutions ($1024 \to 512 \to 1024$) hold about 1.05 million weights.
4. **Inference time.** All mechanisms take 4.47 to 4.76 ms per image. Differences below 0.1 ms are timing noise, so only self-attention is clearly slower. Even that is only 0.18 ms (about 4%) over the baseline, which would not matter much in practice.
5. **Convergence.** Not really. Only self-attention starts faster in epoch 1, and by epoch 20 all models are at 98.4% to 99.4% training accuracy. All of them also dip at epoch 11 when conv5_x is unfrozen and recover within two or three epochs, so attention does not change this either.
6. **Localization.** Not clearly. The Grad-CAM maps of SE, ECA and CBAM cover the same flower regions as the baseline, and the spatial attention map is almost flat. The only visible change is in the roses image, where SE and CBAM put their strongest response on the right flower instead of spreading it over both.
7. **Does complexity always help?** No. Combined attention adds 1.44 million parameters and gets 92.20%, below SE (132160 parameters, 92.56%). Self-attention is the same story, with ten times the parameters of SE and 0.73 points lower accuracy.
8. **Lightweight against heavy.** Yes. ECA adds only 5 parameters and gets 92.56%, the same as SE. Its macro F1 is 92.35% against 92.44%. So on this dataset the fully connected layers in SE are not really needed, and the cheap version does just as well.

21. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

- ResNet50 with ImageNet weights, then global average pooling, dropout 0.3 and a dense layer with 5 outputs.
- Phase 1: backbone frozen, 10 epochs, Adam, learning rate 10^{-3}.
- Phase 2: conv5_x unfrozen (batch norm kept frozen), 10 epochs, learning rate 10^{-4}.
- Batch size 32, seed 109, and the weights of the last epoch are evaluated.
- Augmentation: flip, rotation up to about 14 degrees, 10% zoom, 10% shift and 10% contrast change.

22. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.
The loss is sparse categorical cross-entropy on the logits. For integer labels this is the same as the formula above.

23. Evaluation Metrics

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

Metrics are computed on the 551 test images with macro averaging. Inference time is ms per image on a Tesla T4 (batch of 32, median of three passes).

24. Required Main Result Table

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	92.74%	93.08%	92.47%	92.50%	23 597 957	4.55 ms
Channel Attention	P_3	92.56%	92.81%	92.44%	92.44%	23 730 117	4.47 ms
Spatial Attention	P_3	91.65%	91.87%	91.25%	91.51%	23 598 056	4.50 ms
Channel+Spatial	P_2	91.83%	92.33%	91.13%	91.53%	23 631 368	4.61 ms
Channel+Spatial	P_3	91.11%	91.98%	90.87%	90.91%	23 730 216	4.63 ms
Channel+Spatial	P_4	91.83%	92.30%	91.64%	91.80%	24 124 520	4.63 ms
Self-Attention	P_3	91.83%	91.94%	91.53%	91.68%	24 908 678	4.73 ms
Additional Attention-1 (ECA)	P_3	92.56%	92.63%	92.42%	92.35%	23 597 962	4.58 ms
Additional Attention-2 (CBAM)	P_3	91.65%	91.75%	91.62%	91.46%	23 730 216	4.63 ms

All values are percentages of the 551 test images. Training accuracy is the last-epoch value on augmented images. The best attention rows here are Channel (SE) and ECA at 92.56%, and every attention row is still below the baseline.

25. Graphical Performance Analysis

25.1 Training and Validation Accuracy

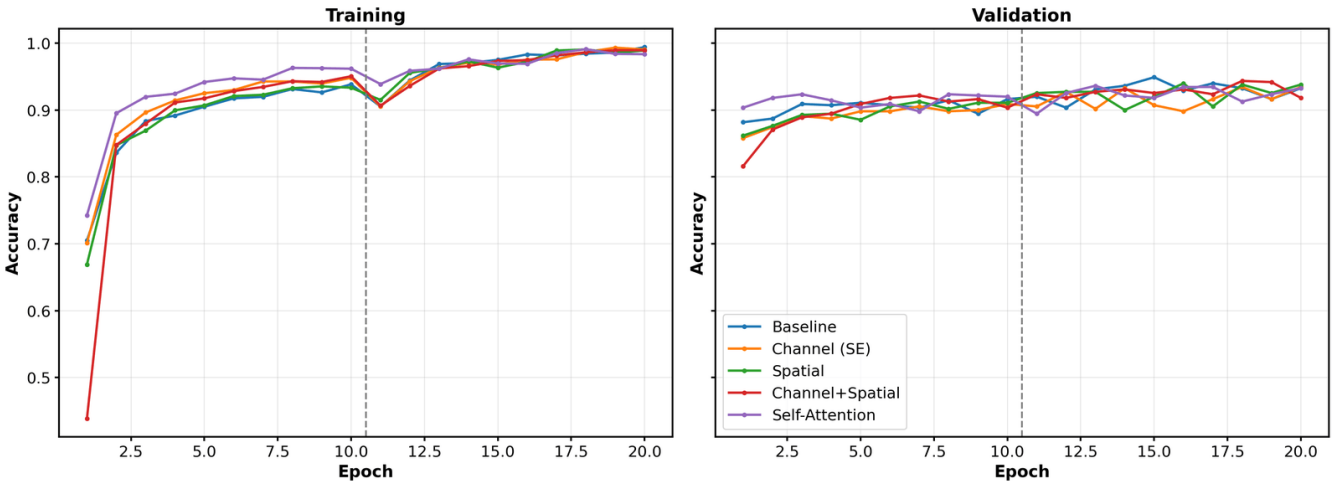


Figure 2: Training (left) and validation (right) accuracy against epoch. The dashed line marks the start of fine-tuning.

Self-attention starts fastest and Channel+Spatial slowest, but all models end near 99% training and 92% to 94% validation accuracy. None is clearly better. Training accuracy dips at epoch 11 when conv5_x is unfrozen and then climbs again, while validation accuracy stays at about 90% to 94% from epoch 3 onwards.

At the last epoch training accuracy is 5 to 7 points above validation accuracy for every model (6.15 points for the baseline, 7.13 for Channel+Spatial), so attention does not close this gap. After fine-tuning all validation curves move up and down by about 2 points, and no attention model is smoother than the baseline.

25.2 Training and Validation Loss

Validation loss stops improving after epochs 12 to 17 and then moves up and down while training loss keeps falling. So the models overfit, and last-epoch values are noisy. For example, Channel+Spatial jumps from about 0.20 at epoch 19 to 0.47 at epoch 20, which is why its last-epoch validation loss is the highest in the tables.

25.3 Attention Mechanism versus Accuracy

The baseline is the highest bar (92.74%) and Channel+Spatial the lowest (91.11%). The axis does not start at zero, so the 1.6 point gap (9 images) looks bigger than it is. The bars fall into two groups: SE and ECA sit just under the baseline at 92.56%, and spatial, CBAM, self-attention and Channel+Spatial sit lower, between 91.11% and 91.83%.

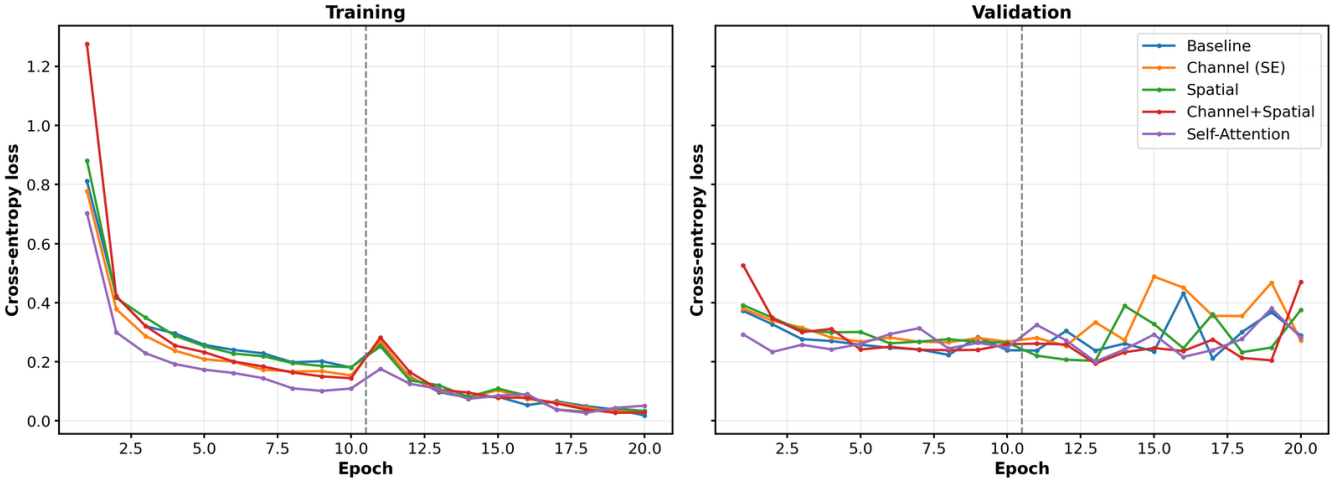


Figure 3: Training (left) and validation (right) loss against epoch. The dashed line marks the start of fine-tuning.

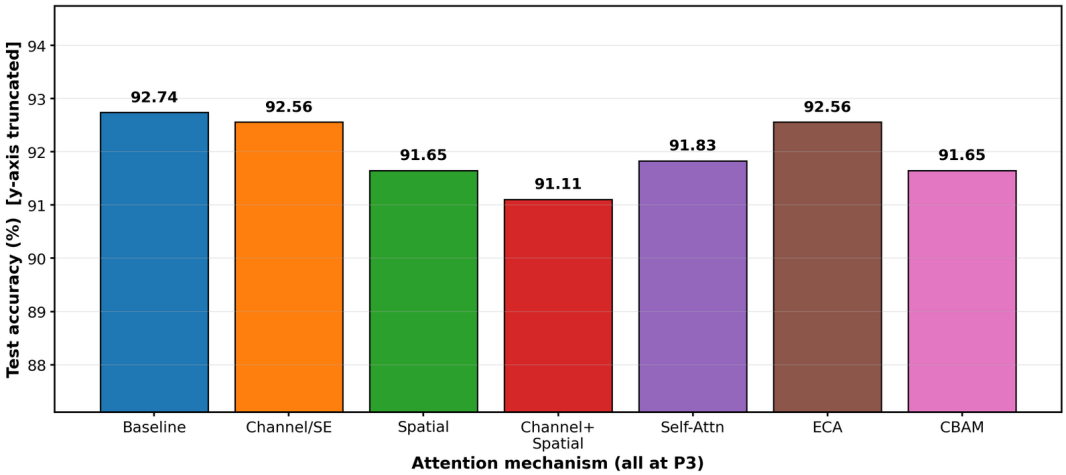


Figure 4: Test accuracy of the attention mechanisms at P_3. The vertical axis does not start at zero.

25.4 Attention Mechanism versus Macro F1

This comparison is useful when the class-wise performance is not uniform.

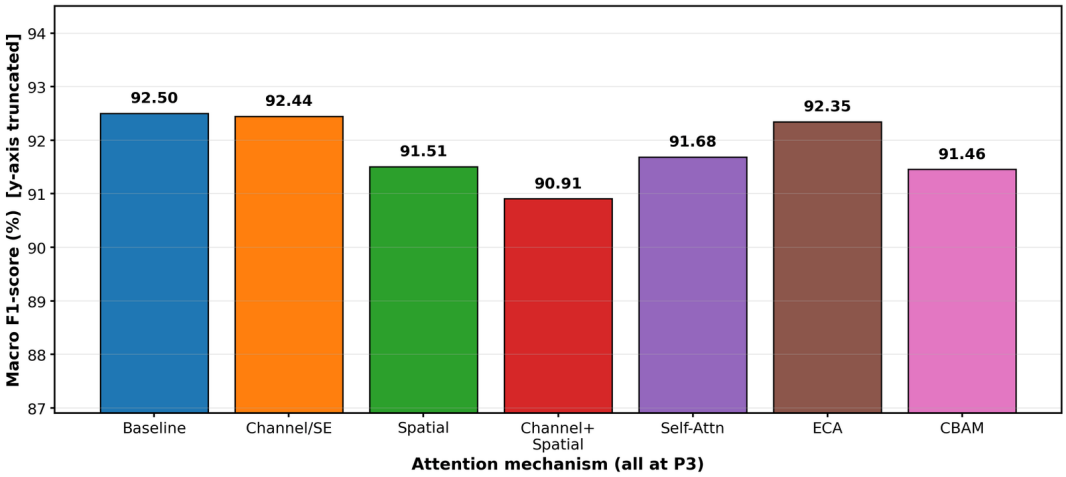


Figure 5: Macro F1-score of the attention mechanisms at P_3. The vertical axis does not start at zero.

Macro F1 gives the same order as accuracy, from 92.50% for the baseline to 90.91% for Channel+Spatial. So the class-wise results do not change the ranking. The two channel-only mechanisms are again closest to the baseline, with SE at 92.44% and ECA at 92.35%.

25.5 Accuracy versus Model Parameters

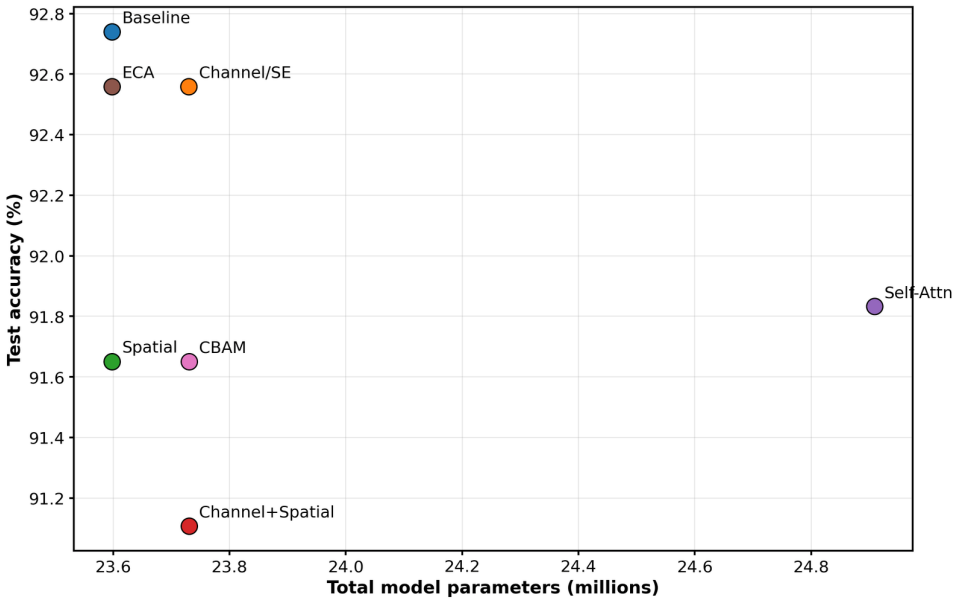


Figure 6: Test accuracy against total parameters at P_3.

Only self-attention has clearly more parameters (24.91 million) and its accuracy is lower than the baseline. More parameters did not give higher accuracy. The other mechanisms all add less than 0.14 million parameters, and among them ECA, which adds almost nothing, ties for the best accuracy.

25.6 Macro F1 versus Inference Time

The times lie between 4.47 and 4.73 ms per image, so the differences are small. Self-attention is the slowest and has a lower F1 than the baseline, so the extra time gave no gain. SE is the fastest model (4.47 ms) and also has the highest F1 among the attention models, so being slower did not mean being better.

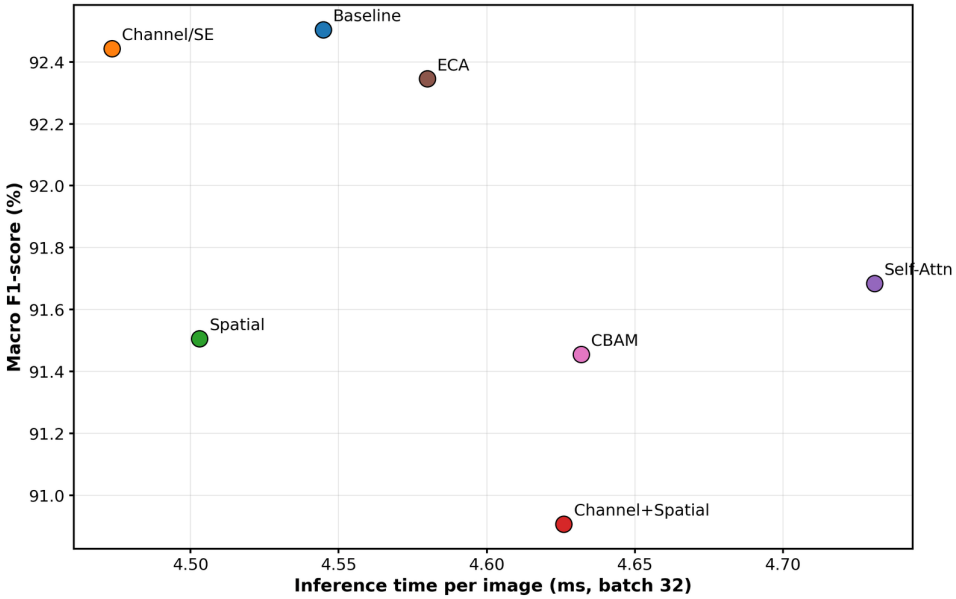


Figure 7: Macro F1-score against inference time per image at P_3.

26. Confusion Matrix Analysis

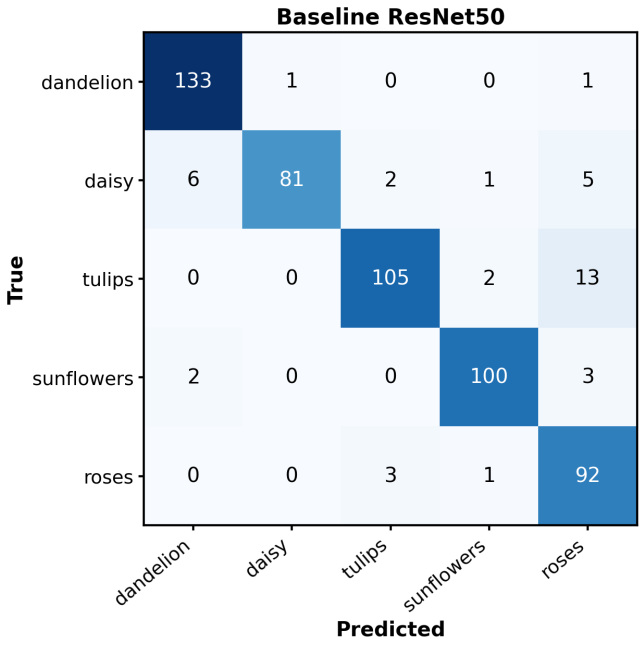
Tulips are most often predicted as roses (13 images in the baseline, 20 with A3). A3 loses 5 tulips and 7 sunflowers and gains 2 roses and 1 dandelion. Most of the lost sunflowers go to dandelion (3 images), probably because both are yellow. The daisy count on the diagonal does not change.

Dandelion and sunflowers are the best classified classes, with baseline F1 of 96.38% and 95.69%. A3 does not reduce any particular confusion. The biggest one, tulips predicted as roses, actually goes up, and the 9 lost images come mostly from tulips and sunflowers, so the change is not spread evenly across classes.

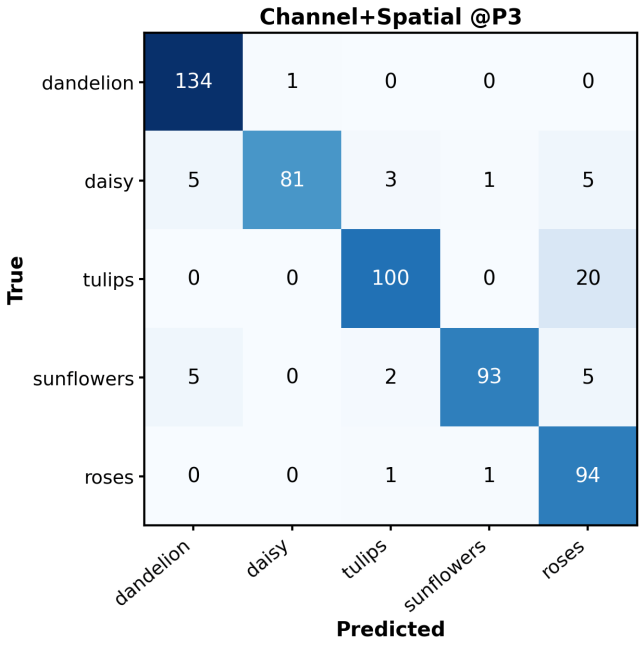
Table 7: Class-wise F1-score (%) for all sixteen runs.

Run	**dandelion**	**daisy**	**tulips**	**sunflowers**	**roses**
A0	96.38	91.53	91.30	95.69	87.62
A1	95.65	93.41	89.08	97.12	86.96
A2	94.93	94.51	87.70	95.73	84.66
A3	96.06	91.53	88.50	93.00	85.45
A4	96.24	92.97	87.65	95.69	85.86
A5	93.24	93.41	90.00	96.19	87.83
B1	93.82	89.13	89.33	94.06	87.23
B2	94.96	92.31	88.98	97.61	83.80
B4	95.02	95.08	87.50	94.93	86.49
M34	96.32	92.31	89.08	96.71	87.31
M234	94.29	91.71	89.06	97.58	83.15
O_SC	97.01	94.57	91.29	97.17	89.34
K3	95.34	92.13	86.53	90.82	88.24
K5	96.70	94.62	90.13	96.19	88.00
ECA	97.04	92.90	90.43	96.19	85.17
CBAM	95.94	92.39	88.89	94.34	85.71

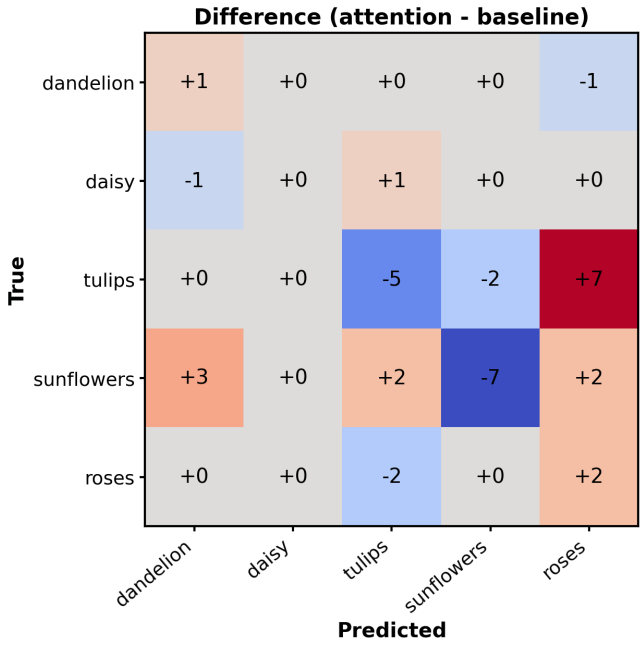
Daisy has a higher F1 than the baseline in 13 of the 15 attention runs, while tulips are lower in all 15. So attention helps daisy but hurts tulips, and the overall accuracy does not improve. Tulips and roses are the two weakest classes in 15 of the 16 runs, which fits the colour overlap seen in the sample images.



(a) Baseline ResNet50



(b) Channel+Spatial at P_3 (A3)



(c) Difference (A3 minus baseline)

Figure 8: Confusion matrices on the 551 test images (counts). Rows are the true classes and columns the predicted classes.

27. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

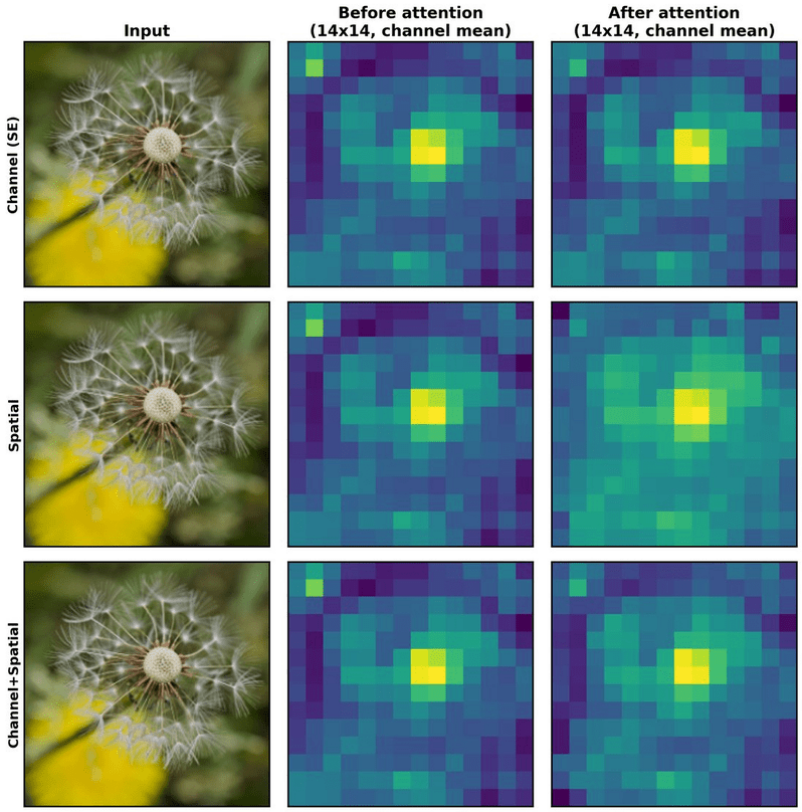


Figure 9: Channel-averaged feature map at P_3 before and after attention (rows: channel, spatial, Channel+Spatial; columns: input, before, after).

The channel mean map looks almost the same before and after attention. The centre of the seed head stays the brightest in all three rows. The spatial and Channel+Spatial rows dim the bright cell in the top-left corner a little, which matches the low border values of the spatial map. Otherwise the attention only rescales the map.

28. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

On the three correct images both models look at the same flower regions. For the last daisy image the attention model looks at the ground below the flower and predicts roses. On the fourth daisy both models are wrong (tulips and roses), and the attention map becomes a narrow band on the petal tips.

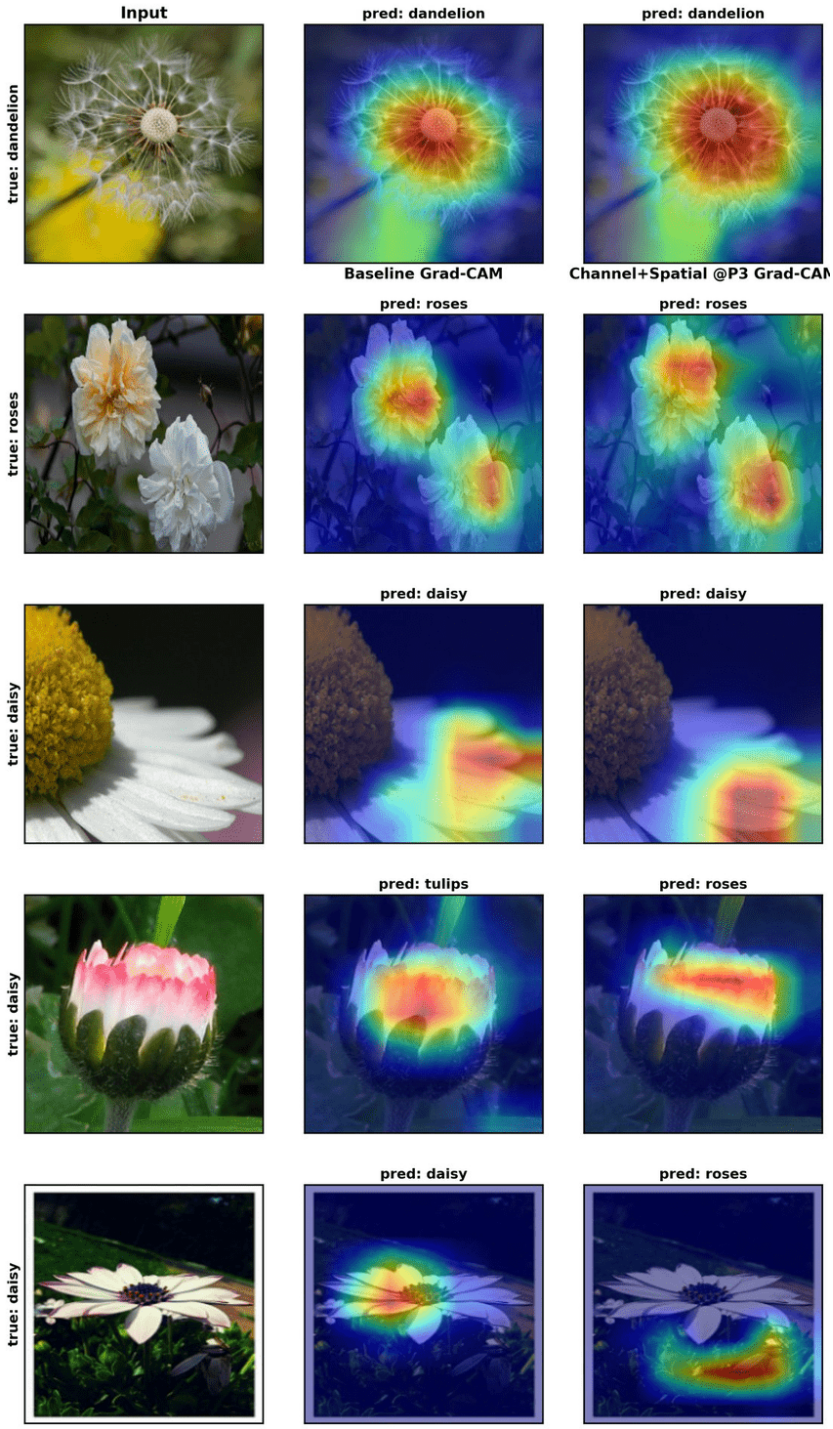


Figure 10: Input, baseline Grad-CAM and Channel+Spatial Grad-CAM for three correct and two incorrect test images (predicted class above each map).

28.1 Grad-CAM Interpretation

- **Focus on the flower:** not stronger. On the correct images both models look at the same regions.
- **Background suppression:** not visible. The baseline already has low activation on the background.
- **Petals:** yes for the daisy images, in both models.
- **Central structure:** both models activate the dandelion seed head, but neither activates the yellow centre of the daisy close-up.
- **Edges:** not visible, as the maps are smooth blobs at 7×7 size.
- **Whole flower:** in the roses image both models cover both flowers.
- **Leaves:** only in the last daisy image, where the attention model's map lies on the ground and plants below the flower. The baseline puts almost no activation on leaves.
- **Background objects:** in the dandelion image both models give weak activation on the blurred yellow flower in the bottom-left corner, and attention does not remove it.
- **Incorrect prediction:** yes. The attention maps change for both incorrect images.

29. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.

An attention module directly computes weights during the forward pass.

Grad-CAM, in contrast, is generated after prediction using gradients.

29.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

About a third of the channels are near 0 and the rest near 1, and the weights are almost the same for every image. So channel attention works like a fixed mask. Very few channels fall between 0.2 and 0.8 in the histogram, so the gate is close to an on/off switch learned during training rather than something that adapts to each flower.

29.2 Spatial-Attention Visualization

For spatial attention:

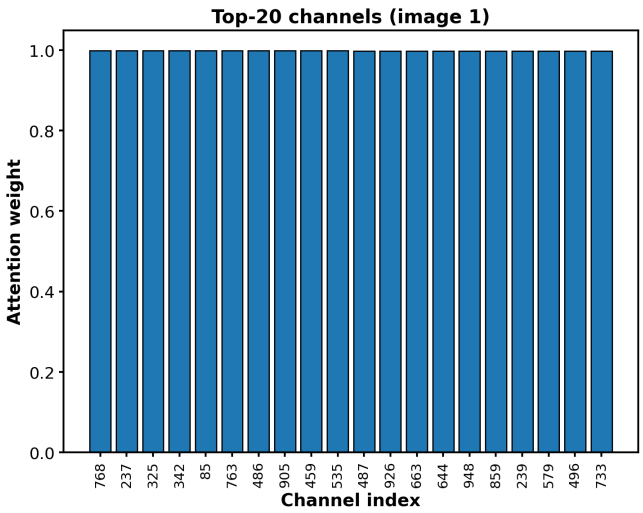
$$M_s(F) \in \mathbb{R}^{H \times W}.$$

The map is flat inside (mean 0.94) and changes only at the image border, so it does not highlight the flower. The low values sit mostly along the top and left edges, and the raw minimum ranges from 0.01 to 0.26 between images. This looks like an effect of zero padding in the 7×7 convolution rather than anything learned about the flower.

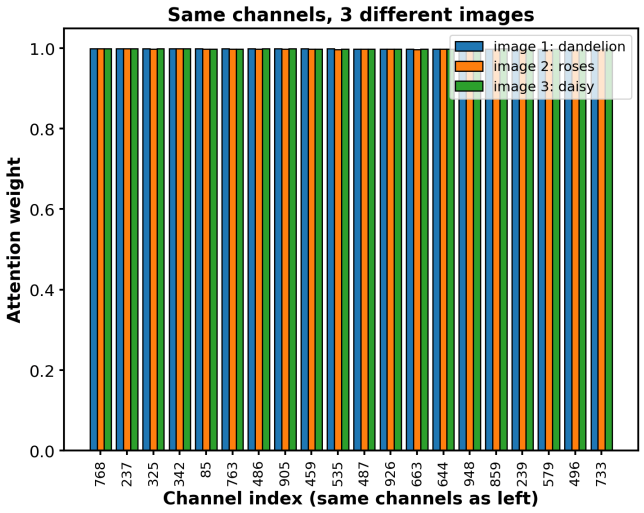
29.3 Self-Attention Visualization

The response is one spot on the flower, the same for every query, and the scale γ is only -0.02. So this module barely changes the features. The mean, maximum and centre-query maps are almost identical, which means every location attends to the same single position. That spot is not always on the most useful part either, for example in the roses image it lies only on the left flower.

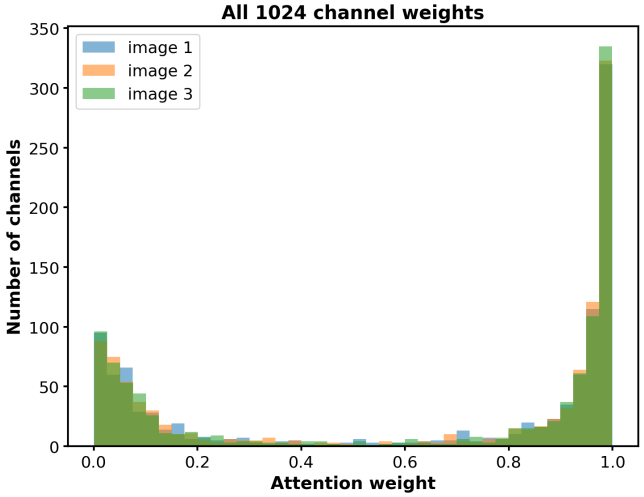
So distant regions do not really interact in this module. A query on the petals and a query on the background both take their features from the same spot, instead of linking related parts of the flower to each other.



(a) Top-20 channels of image 1



(b) Same channels, three images



(c) All 1024 channel weights

Figure 11: Channel-attention weights of the channel (SE) model at P_3. Images 1 to 3 are the dandelion, roses and daisy test images.



Figure 12: Spatial attention (A2, P_3) for the five test images: input, map (scaled to its own minimum and maximum) and overlay. The raw range is written under each map.

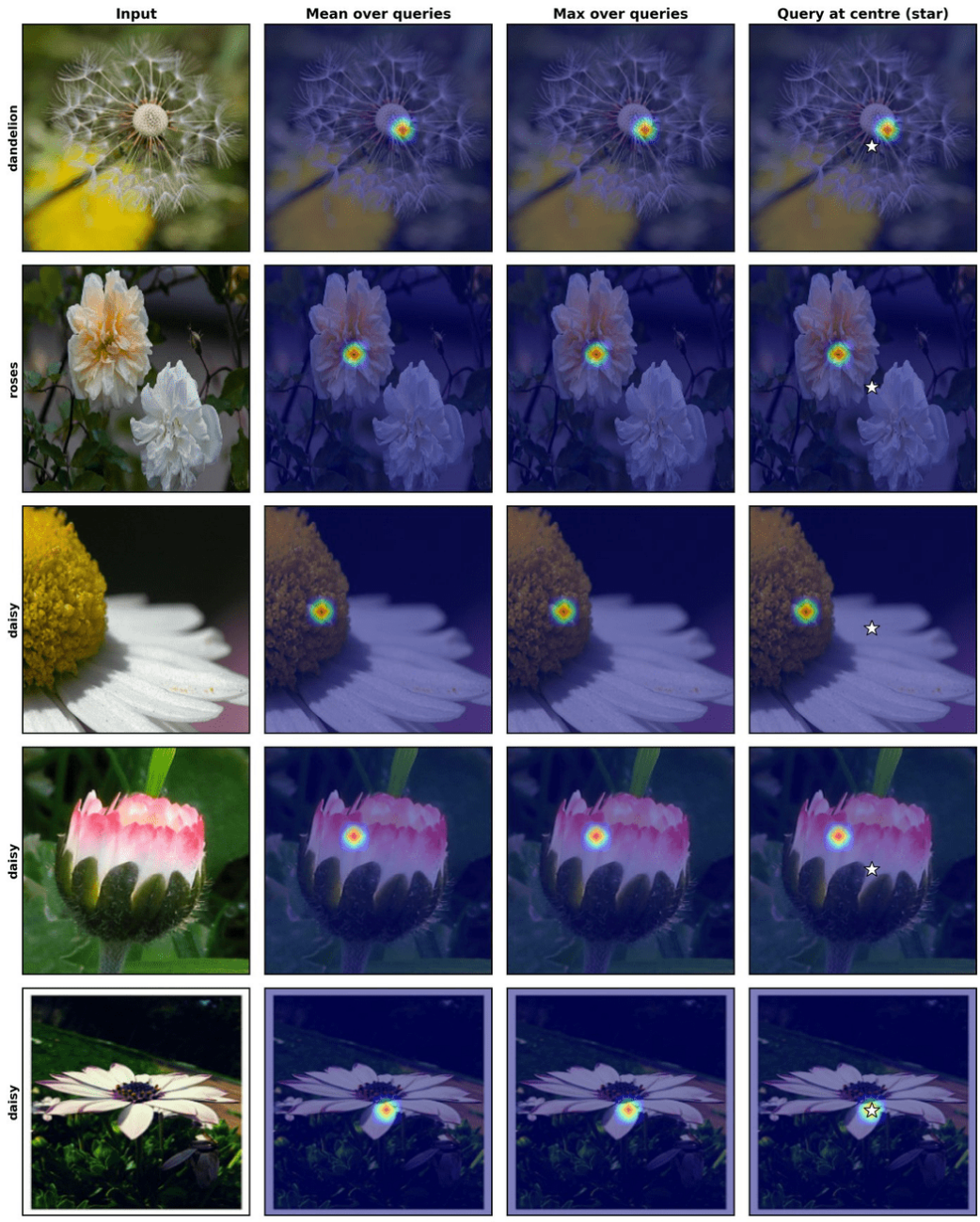


Figure 13: Self-attention response of A4 at P_3: input, mean over queries, maximum over queries and the query at the centre (star).

30. Attention Position Visualization

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

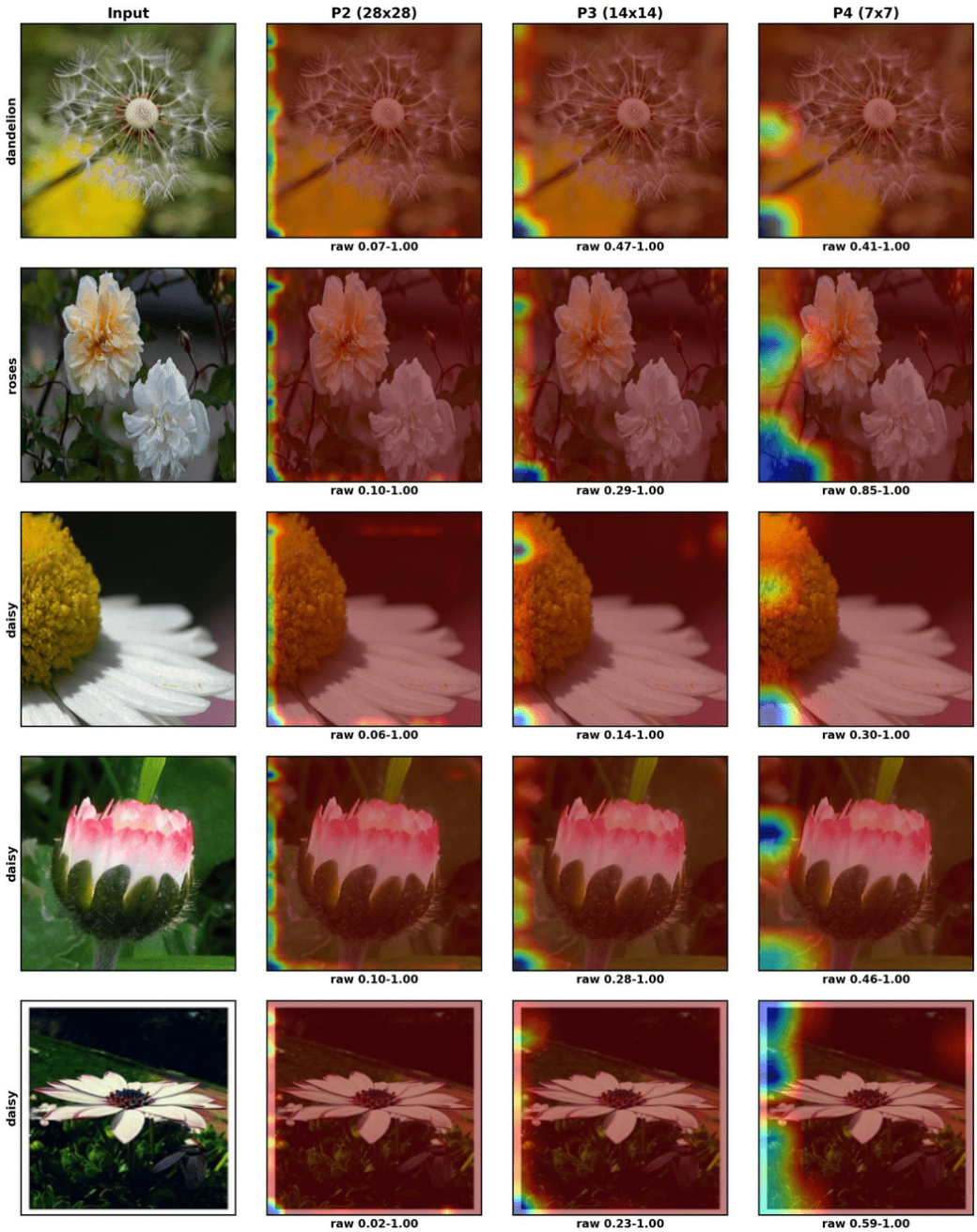


Figure 14: Spatial attention maps at P_2, P_3 and P_4 for the five test images (models B2, A3 and B4). Raw minimum values are written under each map.

Table 8: Statistics of the spatial attention maps over 100 test images.

Model (position)	**Mean value**	**Minimum**	**Locations above 0.5**
B2 (P_2)	0.978	0.002	98.4%
A3 (P_3)	0.983	0.093	99.3%
B4 (P_4)	0.973	0.008	97.9%

The maps at P_2, P_3 and P_4 look alike, with a flat interior and low values only at the border. So no change from edges to flower parts is seen with depth. The table agrees: at all three positions the mean is about 0.97 to 0.98 and over 97% of locations are above 0.5, so almost everything passes through. Only the border blocks get bigger.

31. Qualitative Comparison Across Attention Mechanisms

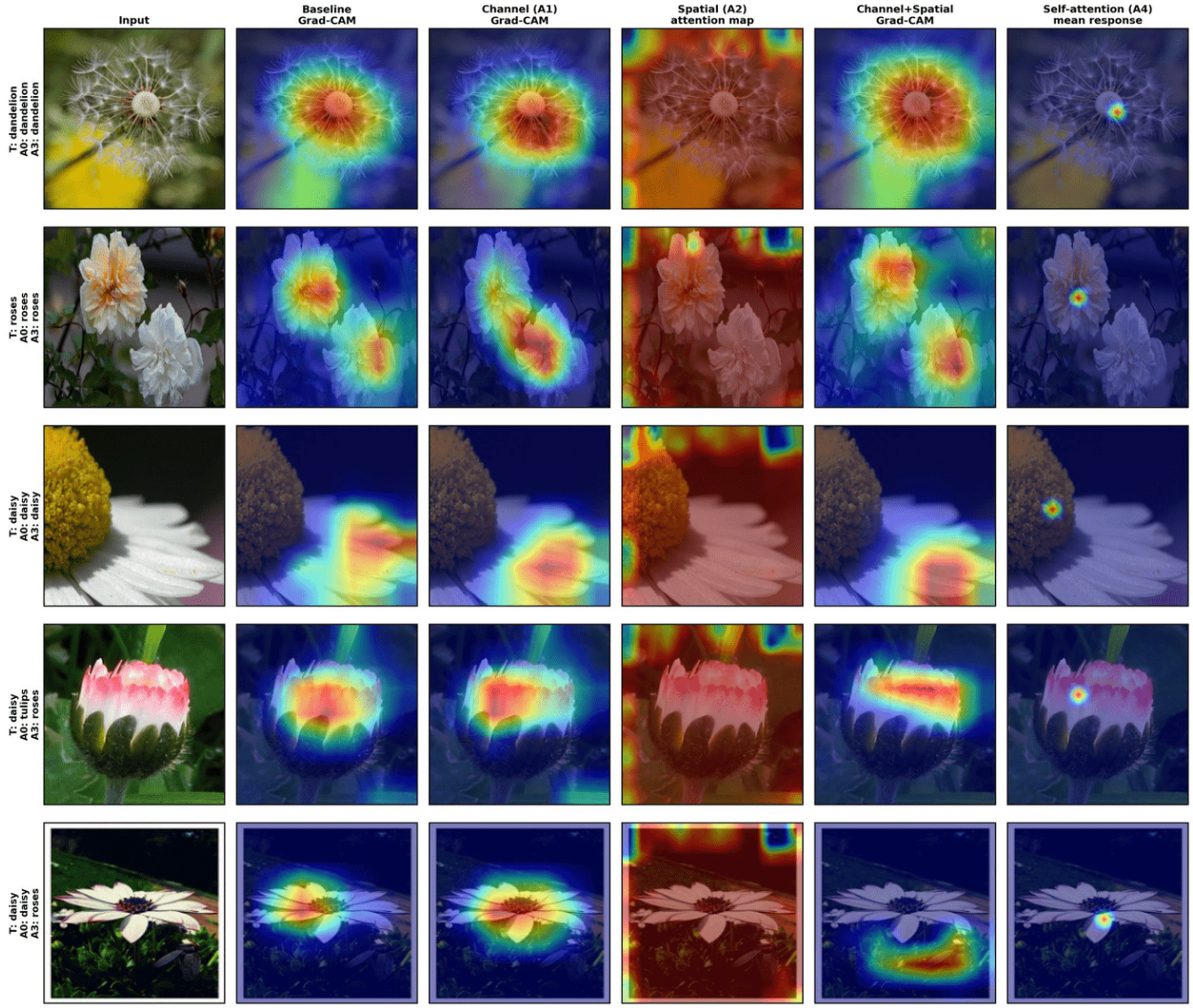


Figure 15: Three correct and two incorrect test images. Columns: input (T = true class, A0 = baseline prediction, A3 = Channel+Spatial prediction), baseline Grad-CAM, channel (A1) Grad-CAM, spatial (A2) map, Channel+Spatial (A3) Grad-CAM, self-attention (A4) response.

The maps look similar for the models on the correct images and differ on the two wrong ones. The spatial map is flat and the self-attention map is one spot. On the correct images even the A1 Grad-CAM is close to the baseline, except for roses where it joins both flowers. Both models miss image 4 and only A3 misses image 5.

31.1 Qualitative Criteria

- **Foreground and background:** the Grad-CAM maps lie on the flower in all images except the last one for A3, and no background suppression is visible.
- **Petals and centre:** the daisy maps lie on the petals, and the yellow centre of the close-up gets no activation.
- **Compactness:** in the daisy close-up the A3 map is a compact block, and in the roses image the SE map is one region joining both flowers.
- **Activation coverage:** the baseline and A1 maps cover most of the flower in the dandelion and daisy images, while the A3 map on the fourth image covers only a thin band of the petals.
- **Incorrectly attended regions:** only the last image, where the A3 map lies on the ground below the flower.

- **Correct against incorrect:** on the three correct images every Grad-CAM lies on the flower, while on the two wrong ones the A3 map shrinks to a band or moves off the flower.
- **Consistency and depth:** the spatial maps show the same border pattern in every image and at every position, and the self-attention response is always one spot.

32. Required Visualizations

1. Sample input images from all five classes;
2. Training and validation accuracy curves;
3. Training and validation loss curves;
4. Confusion matrix;
5. Feature map before attention;
6. Channel-attention weights;
7. Spatial-attention heatmap;
8. Spatial-attention overlay;
9. Self-attention response for the self-attention experiment;
10. Feature map after attention;
11. Baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. Attention maps at P_2, P_3, and P_4; and
14. At least one graphical comparison of the performances of different attention mechanisms.

Where each required visualization is shown

Table 9: Required visualizations and the figures that contain them.

No.	Required visualization	Figure
1	Sample input images from all five classes	1
2	Training and validation accuracy curves	2
3	Training and validation loss curves	3
4	Confusion matrix	8
5	Feature map before attention	9
6	Channel-attention weights	11
7	Spatial-attention heatmap	12
8	Spatial-attention overlay	12
9	Self-attention response	13
10	Feature map after attention	9
11	Baseline Grad-CAM	10
12	Grad-CAM of an attention-based model	10, 15
13	Attention maps at P_2, P_3 and P_4	14
14	Comparison of attention mechanisms	4, 5, 17

33. Recommended Visualization Layout

SE, ECA and CBAM cover the same flower regions as the baseline with small shifts, and none of them activates the background. ECA looks the most like the baseline in all three images. In the roses image CBAM puts its peak on the right flower and SE stretches one region over both, while self-attention again gives one small spot.

34. Required Experimental Analysis

34.1 Quantitative Analysis

Accuracy + Precision + Recall + Macro F1

Thirteen of the fifteen attention runs are below the baseline. The two above it (94.01% and 93.28%) are not supported by neighbouring settings. The rest lie between 90.74% and 92.56%, which is smaller than the run-to-run noise in validation accuracy, so none of those settings really stands out.

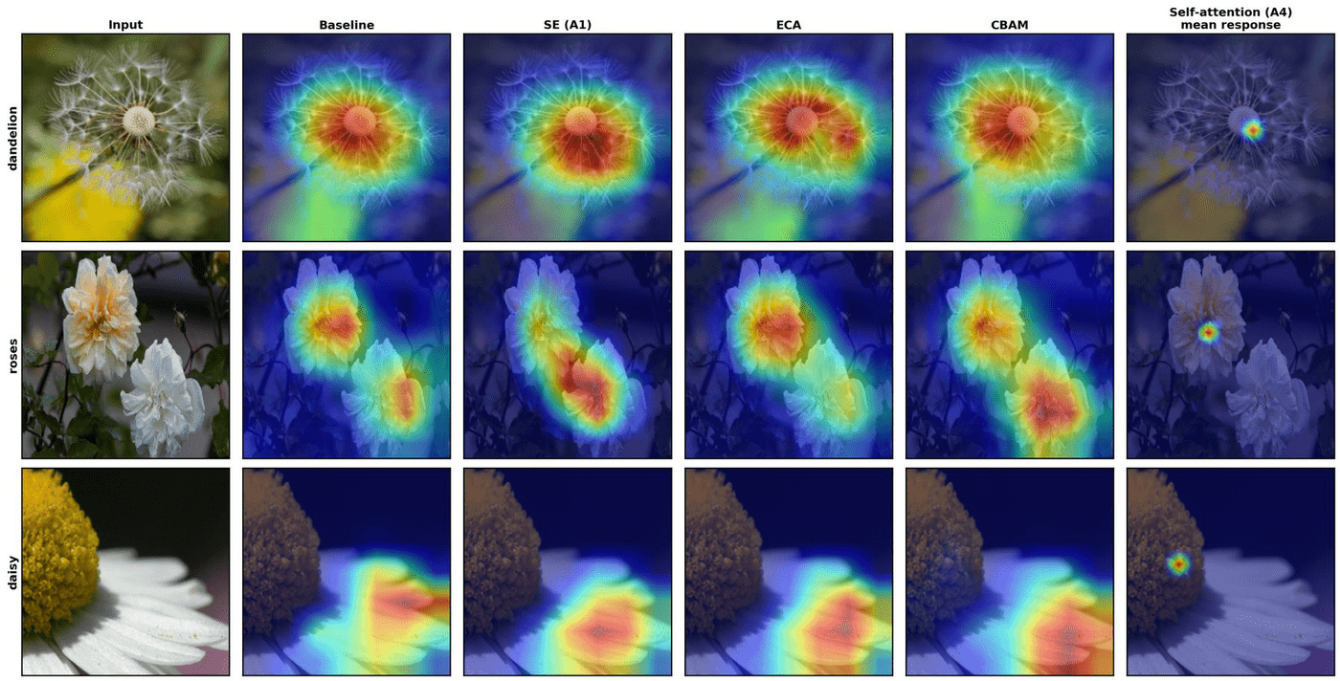


Figure 16: Three correct images. Columns: input, baseline, SE (A1), ECA, CBAM and self-attention (A4) response.

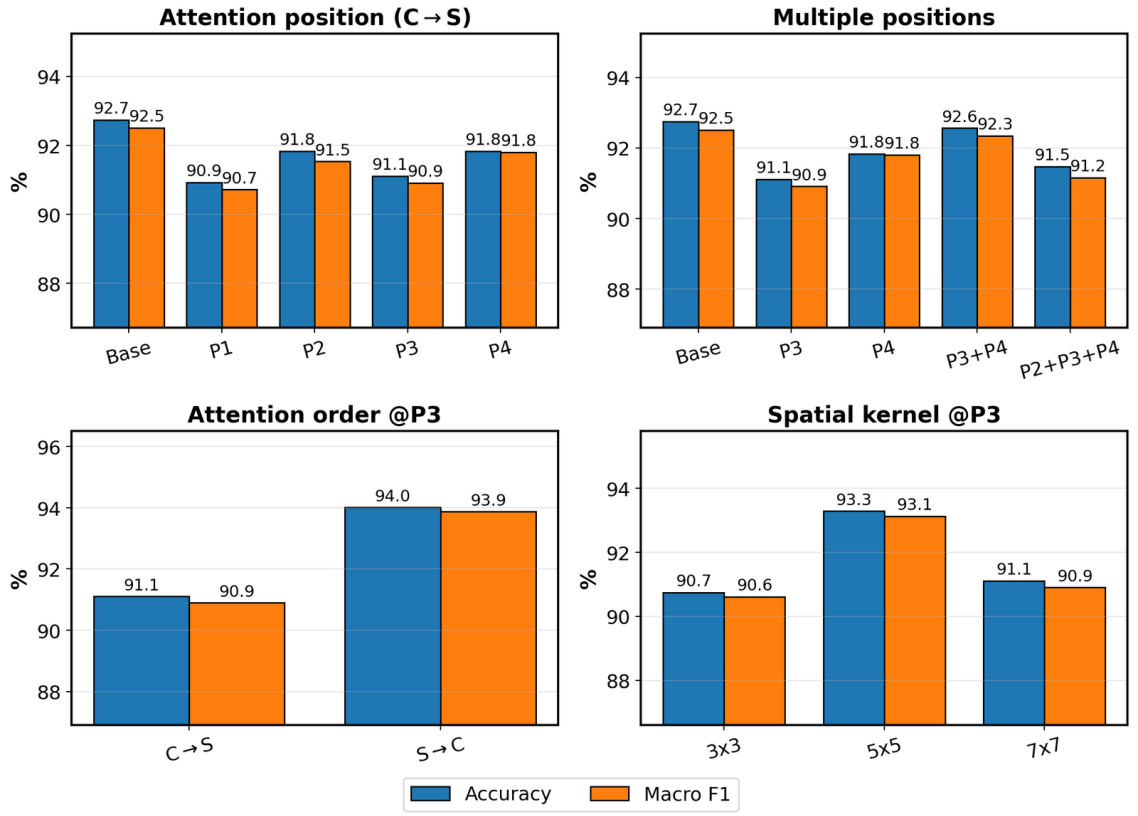


Figure 17: Accuracy and macro F1 for the position, multiple-position, order and kernel-size experiments. The vertical axes do not start at zero.

34.2 Complexity Analysis

$$\boxed{\text{Parameters} + \text{Inference Time}}$$

Added parameters range from 5 (ECA) to 1.44 million (combined attention). Inference time stays between 4.47 and 4.76 ms per image, and only self-attention and combined attention are slower, by about 0.2 ms. Even the largest module adds only about 6% to the 23.6 million parameters of ResNet50, so cost is not what decides between these mechanisms here.

34.3 Qualitative Analysis

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Therefore, the overall evaluation should be:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The Grad-CAM maps of the attention models match the baseline on the correct images and change on the misclassified ones. The internal maps are not informative: flat, almost fixed, or one spot. So the qualitative results agree with the numbers, since the attention models neither focus better nor score better than the baseline.

35. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

All sixteen runs use the same split, preprocessing, optimizer and epochs, and only the attention module changes. Each setting was trained once (seed 109). Validation accuracy moves 2.8 points on average over the last six epochs, so gaps of about 2 points are treated as noise. Because of this, close runs are not ranked in the text.

Table 10: The sixteen training runs.

Run	Attention module	Position
A0	Baseline	–
A1	Channel (SE)	P_3
A2	Spatial (7×7)	P_3
A3 (= B3)	Channel→Spatial (7×7)	P_3
A4	Self-attention	P_3
A5	Channel+Spatial+Self-attention	P_3
B1	Channel→Spatial	P_1
B2	Channel→Spatial	P_2
B4	Channel→Spatial	P_4
M34	Channel→Spatial	$P_3 + P_4$
M234	Channel→Spatial	$P_2 + P_3 + P_4$
O_SC	Spatial→Channel	P_3
K3	Channel→Spatial, 3×3 kernel	P_3
K5	Channel→Spatial, 5×5 kernel	P_3
ECA	Efficient Channel Attention	P_3
CBAM	CBAM	P_3

36. Minimum Mandatory Ablation Set

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

All ten items of the minimum set were completed (A0 to A4, B2, B4, the Grad-CAM comparison, the P_2, P_3, P_4 maps and the comparison graphs).

37. Additional Exploration

ECA and CBAM were added at P_3 with the same protocol. ECA gets 92.56% with 5 extra parameters and CBAM gets 91.65% with 132259. CBAM ends up level with plain spatial attention, so its shared channel network did not add anything on top of the spatial gate.

38. Questions and Answers

1. What is the purpose of residual connections in ResNet50?
   Answer. Residual connections add the block input to its output, $y = F(x) + x$. This gives gradients a direct path to earlier layers, so the 50-layer network can be trained without accuracy dropping as depth grows. Each block only has to learn a change to its input, and can stay close to the identity if nothing useful is learned.
2. What does conv2_x represent compared with conv5_x?
   Answer. conv2_x is the first stage (56×56, 256 channels) and learns edges and textures. conv5_x is the last stage (7×7, 2048 channels) and learns class-level features. Each cell of conv5_x corresponds to a 32×32 patch of the input against 4×4 for conv2_x, so it sees much more of the flower at once.
3. Why does the spatial resolution decrease with network depth?
   Answer. The first block of conv3_x, conv4_x and conv5_x uses stride 2. This halves the height and width, which increases the receptive field and reduces computation. Without it the deeper layers, which have many more channels, would be far too expensive to run on a 56×56 map.
4. Why does the number of feature channels increase with network depth?

Answer. The channels double at each stage so the network can store more combinations of patterns, which makes up for the lost spatial detail. Since the height and width are halved at the same time, the cost per layer stays roughly the same from stage to stage.

5. What is meant by low-level, intermediate, and high-level features?
   Answer. Low-level features are edges, colours and simple textures. Intermediate features are petals and repeated shapes. High-level features describe the whole flower and its class. In ResNet50 these roughly match conv2_x, then conv3_x and conv4_x, and then conv5_x, although the boundary between them is not sharp.
6. What is the purpose of an attention mechanism?
   Answer. Attention lets the network give more weight to useful features and less to others, depending on the input. In a CNN this means scaling some channels or locations up and others down before the next layer. The idea is that background and irrelevant patterns then have less influence on the prediction.
7. What does channel attention attempt to learn?
   Answer. Channel attention learns which feature channels are important and gives one weight per channel. Since each channel responds to a certain kind of pattern, this decides what to look for in the image. The same weight is used at every location of that channel.
8. What does spatial attention attempt to learn?
   Answer. Spatial attention learns which locations are important and gives one weight per location. It is built from the mean and maximum over channels, so it decides where to look rather than what to look for. The same weight is applied to all channels at that location.
9. How is channel attention different from spatial attention?
   Answer. Channel attention gives a vector of size C and ignores position. Spatial attention gives an $H \times W$ map and ignores which channel a value comes from. So one reweights feature types and the other reweights image regions, and at P_3 that is 1024 weights against 196.
10. Why can channel and spatial attention be used sequentially?
    Answer. They answer different questions (what and where) and keep the tensor size the same, so the second can be applied on the output of the first. Doing them one after the other is also much cheaper than learning a separate weight for every channel and location together.
11. Why might channel attention behave differently at conv3_x and conv5_x?
    Answer. The channels mean different things at the two stages (edges and parts at conv3_x, class features at conv5_x). But here both positions gave the same accuracy (91.83%). Also, the channel gate at P_3 was nearly fixed, and if that holds at other stages it acts more like a mask than an input-dependent choice.
12. Why can spatial attention be useful at an intermediate ResNet stage?
    Answer. At an intermediate stage the map still has enough resolution (28×28 or 14×14) to find parts. But here the spatial map was almost flat, so no benefit was seen. At P_4 there are only 49 locations, so a spatial map there can only mark very coarse regions.
13. Why is self-attention computationally expensive when the feature map has high spatial resolution?
    Answer. The attention matrix has $N \times N$ entries with $N = H \times W$, so the cost grows with the square of the number of locations. Doubling the height and width makes N four times larger and the matrix sixteen times larger. That is why self-attention was only placed at P_3 in this experiment.
14. What happens to the size of the self-attention matrix when the spatial resolution increases?
    Answer. It grows with the square. It has 2401 entries at 7×7, 38416 at 14×14, 614656 at 28×28 and 9834496 at 56×56. At P_1 that is almost 10 million values per image, or over 300 million for a batch of 32, which is not practical on a single GPU.
15. How does attention performance vary when the insertion position is changed?
    Answer. No position beat the baseline (92.74%). B1 to B4 gave 90.93%, 91.83%, 91.11% and 91.83%, which is a spread of 5 images. There is no trend with depth either, as the middle position P_3 is lower than both P_2 and P_4.
16. How does attention at P_2 differ visually from attention at P_4?
    Answer. Both maps are flat inside and low only near the border. At P_4 the low blocks are larger, as one cell covers 32×32 pixels against 8×8 at P_2. Neither shows the expected change from edges at P_2 to whole flower regions at P_4.
17. Does applying attention at multiple stages always improve classification performance?
    Answer. No. $P_3 + P_4$ gave 92.56%, better than P_3 or P_4 alone but below the baseline, and $P_2 + P_3 + P_4$ dropped to 91.47%. Each extra module adds parameters and one more place where the features get rescaled, and here that did not pay off.
18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?
    Answer. The accuracy differs (91.11% against 94.01%), but the gap is as large as the variation between kernel sizes, so it is not strong evidence for the order. One possible reason for a real difference is that in Spatial→Channel the channel weights are computed after the border is suppressed, but more seeds are needed to check it.
19. How does the spatial-attention kernel size influence the attention map?
    Answer. The kernel sets how many neighbours are used for each map value. Accuracy was 90.74%, 93.28% and 91.11% for 3, 5 and 7 with no trend, and only the 7×7 map was visualised. At P_3 a 7×7 kernel already covers

half of the 14×14 map, so a larger kernel should give smoother maps, but this was not checked.

20. How does SE attention differ from ECA attention?
    Answer. SE uses two fully connected layers over all channels (132160 parameters). ECA uses a small 1D convolution (kernel 5), so each channel sees only its neighbours (5 parameters). ECA also skips the channel reduction step, and in these runs both got the same accuracy of 92.56%.
21. How does CBAM combine channel and spatial attention?
    Answer. CBAM applies channel attention using average and max pooling, then spatial attention with a 7×7 convolution. In these runs it gave 91.65%. Both pooled vectors pass through one shared network, which is why it has the same parameter count as Channel+Spatial (132259).
22. How does self-attention differ from channel attention?
    Answer. Self-attention compares every location with every other one ($N \times N$ matrix) and adds 1310721 parameters. Channel attention only rescales channels and adds 132160. So self-attention mixes information across the image, while channel attention leaves each location's features in place and only changes their scale.
23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?
    Answer. No. Self-attention has the most parameters and gets 91.83%, while SE and ECA get 92.56% with far fewer. The baseline, with no added parameters at all, is still the best of the P_3 models.
24. What is the difference between an internal attention map and Grad-CAM?
    Answer. An internal attention map is computed by the module in the forward pass and multiplied into the features. Grad-CAM is made after the prediction from gradients. Here the attention map was flat but Grad-CAM was on the flower. So the attention map shows what the module does to the features, while Grad-CAM shows which regions the final prediction depends on.
25. Why is Grad-CAM called a post-hoc explanation technique?
    Answer. It is computed after the model has predicted, using gradients of the trained network, and it does not change the prediction. It can be applied to any CNN without retraining, which is why it was used to compare the baseline and the attention models on equal terms.
26. Does the attention model suppress irrelevant background regions?
    Answer. Not visibly. The spatial map is flat inside, and the attention Grad-CAMs do not show lower background activation than the baseline. The baseline Grad-CAM was already low on the background, so there was not much left for attention to suppress.
27. Are the attention maps of correctly classified and incorrectly classified images different?
    Answer. Yes, the Grad-CAM of the Channel+Spatial model changed for the two wrong images. The internal maps look alike. This is based on only five images. In one wrong image the map became a narrow band on the petal tips, and in the other it moved off the flower to the ground.
28. Which flower classes benefit most from attention?
    Answer. Daisy gains (F1 above the baseline in 13 of 15 attention runs), while tulips lose in all 15, mostly because more tulips are predicted as roses. Dandelion and sunflowers go up or down depending on the run, and roses are lower than the baseline in 11 of the 15 runs.
29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?
    Answer. It shows which classes are confused and in which direction. A3 loses 9 images overall, from sunflowers (-7) and tulips (-5) against small gains elsewhere, which accuracy alone hides. It also shows that the main error is tulips predicted as roses, rising from 13 to 20 images, which points to the colour overlap between them.
30. Why is macro F1 useful in addition to accuracy?
    Answer. Macro F1 gives every class equal weight, so a weak class like roses counts fully. Here it is 0.1 to 0.25 points below accuracy. The gap is small because the test set is close to balanced (95 to 135 images per class), so in this experiment both metrics rank the models almost the same way.
31. What is meant by the accuracy–complexity trade-off?
    Answer. It is the accuracy gained against the parameters and computation added. Self-attention adds 1.31 million parameters and loses 0.91 points, while ECA adds 5 and loses 0.18. Since no module improved accuracy here, the cheapest one (ECA) is the sensible choice if attention has to be used.
32. What is meant by the F1–inference-time trade-off?
    Answer. It is the F1 gained against the extra inference time. Self-attention is the slowest (4.73 ms) and has a lower F1 than the baseline, so the time gave no gain. Among the P_3 mechanisms, SE is the fastest at 4.47 ms and still has the best F1 after the baseline.
33. Why must every ablation study use the same dataset split?
    Answer. So that every model is tested on the same images. Otherwise a difference could come from easier or harder test images, not the model. With 551 test images one image is 0.18 points, so even a few swapped images could change the order of runs that are this close.
34. Why must the same training protocol be maintained when comparing different attention mechanisms?
    Answer. So that only the attention changes. If epochs or learning rates differ, the difference could come from training and not from the attention. Here the start of fine-tuning alone causes a visible dip and recovery in the curves, so changing when it starts would already move the results.
35. What conclusions can be drawn from the Grad-CAM comparison?

Answer. On five images, the attention model looks at the same flower regions on the correct ones, and its map changes on the wrong ones. There is no sign that it localizes better. Five images is a small sample, so this only shows that attention did not obviously help here, not that it never could.

39. Observations to be Recorded

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

Observation 1 is the dimension table. Observations 2 to 12 are in the table below for all sixteen runs. Observations 13 to 18 are the confusion matrices, attention maps, Grad-CAM and position maps shown in the figures.

Table 11: Recorded values of all sixteen runs. Losses are from the last epoch; inference time is in ms per image.

Run	Train Acc	Val Acc	Test Acc	Train Loss	Val Loss	Prec.	Recall	F1	Trainable Params	Total Params	Time
A0	99.42	93.27	92.74	0.018	0.289	93.08	92.47	92.50	14 963 717	23 597 957	4.55
A1	99.07	93.45	92.56	0.033	0.272	92.81	92.44	92.44	15 095 877	23 730 117	4.47
A2	98.91	93.82	91.65	0.033	0.376	91.87	91.25	91.51	14 963 816	23 598 056	4.50
A3 / B3	98.95	91.82	91.11	0.029	0.470	91.98	90.87	90.91	15 095 976	23 730 216	4.63
A4	98.37	93.27	91.83	0.051	0.283	91.94	91.53	91.68	16 274 438	24 908 678	4.73
A5	98.40	93.82	92.20	0.046	0.363	92.57	91.83	92.13	16 406 697	25 040 937	4.76
B1	98.05	92.73	90.93	0.057	0.278	91.26	90.39	90.71	14 972 280	23 606 520	4.69
B2	98.17	92.55	91.83	0.049	0.280	92.33	91.13	91.53	14 997 128	23 631 368	4.61
B4	98.48	92.91	91.83	0.039	0.315	92.30	91.64	91.80	15 490 280	24 124 520	4.63
$P_3 + P_4$	98.21	94.00	92.56	0.068	0.244	92.50	92.29	92.35	15 622 539	24 256 779	4.66
$P_2 + P_3 + P_4$	98.64	93.09	91.47	0.034	0.300	92.13	90.68	91.16	15 655 950	24 290 190	4.72
S→C	98.13	94.91	94.01	0.065	0.218	93.96	93.86	93.88	15 095 976	23 730 216	4.61
$k = 3$	97.39	92.91	90.74	0.075	0.295	91.42	90.34	90.61	15 095 896	23 730 136	4.64
$k = 5$	98.17	93.27	93.28	0.057	0.245	93.22	93.15	93.13	15 095 928	23 730 168	4.56
ECA	98.48	92.55	92.56	0.049	0.312	92.63	92.42	92.35	14 963 722	23 597 962	4.58
CBAM	98.68	91.82	91.65	0.045	0.391	91.75	91.62	91.46	15 095 976	23 730 216	4.63

40. Discussion Guidelines

Why the performance changed. Attention did not help here. The spatial gate is close to 1 everywhere, the channel gate is nearly fixed and the self-attention scale is only about -0.02, so the modules change the features very little. A likely reason is that the pretrained ResNet50 features are already good for flowers, and with 2569 training images and 20 epochs the modules had little to learn on top of them.

Location and resolution. Position made no clear difference (90.93% to 91.83%). The maps at P_2, P_3 and P_4 look alike, so the expected move from edges to semantic regions was not seen. Since the gate is close to 1 at every position, it does not matter much where it is placed, which fits the small spread in accuracy.

Complexity. Extra parameters did not buy accuracy. ECA (5 parameters) and SE (132160) both get 92.56%, while self-attention (1.31 million) gets 91.83%. Time differences are under 0.3 ms. Most of the run time is spent in the backbone, so even the heaviest module adds only about 5% to the inference time.

Maps and background. The attention maps do not match object regions and the background is not suppressed. Grad-CAM does lie on the flower, but it looks the same as the baseline. So the flower localization comes from the pretrained backbone, not from the added attention.

Incorrect predictions. For the two images misclassified by Channel+Spatial, the Grad-CAM changed from the baseline, and in one it moved to the ground below the flower. This is only two images. Both are daisies predicted as roses, which matches the confusion matrix, where A3 sends more images of other classes to roses.

Quantitative against qualitative. Both agree that attention added nothing. The only consistent pattern is by class: daisy improves and tulips get worse. Each setting was trained once, so small gaps are noise. Running each setting with three or more seeds would show whether the daisy and tulip trend, and the high Spatial→Channel result, are real.

41. Expected Outcome

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

42. Conclusion

Sixteen ResNet50 models with different attention types and positions were trained on the same flowers split. The baseline got 92.74% and no attention model beat it clearly.

The modules were almost inactive (flat spatial gate, fixed channel gate, self-attention scale near zero), and gaps of about 2 points are within noise. Repeating with several seeds would be needed to confirm smaller differences.

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